Service Desk

ITIL®4 Practice Guide

AXELOS

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CHAPTER 1

ABOUT THIS DOCUMENT

1 About this document

This document provides practical guidance for the service desk practice. It is split into seven main sections, covering:

- general information about the practice
- the practice's processes and activities and their roles in the service value chain
- the organizations and people involved in the practice
- the information and technology supporting the practice
- considerations for partners and suppliers for the practice.
- information on assessing and developing the capability of the practice
- recommendations for succeeding in the practice.

1.1 ITIL® 4 qualification scheme

Selected content from this document is examinable as a part of the following syllabuses:

- **ITIL Specialist** Create, Deliver and Support
- **ITIL Specialist** Drive Stakeholder Value
- **ITIL Specialist** Service Desk
- **ITIL Specialist** Monitor, Support, and Fulfil

Please refer to the respective syllabus documents for details.

CHAPTER 2

GENERAL INFORMATION

2 General Information

2.1 Purpose and description

Key message

The purpose of the service desk practice is to capture demand for incident resolution and service requests. It should also be the entry point and single point of contact for the service provider for all users.

Note: In some organizations, the main purpose of the service desk practice is establishing an effective communication interface between a service provider and its users, with incidents and service requests being just two subjects of communication. In these organizations, the purpose of this practice could be to establish an effective entry point and single point of contact with all users and to capture demand for incident resolution and service requests. Organizations can and should adjust the practice purpose statements and the other recommendations of ITIL according to their objectives and circumstances.

As any practice, this practice includes all four dimensions of service management.

Table 2.1 Dimension of service management with examples

Dimension of service management	Example of service desk practice resources
Organizations and people	Dedicated team, sometimes known as the service desk
Information and technology	Dedicated information system, sometimes known as the service desk
Value streams and processes	Workflows and procedures for communications with users and other teams
Partners and suppliers	Involved third parties, in some cases known as the service desk

The term ‘service desk’ can refer to various types and groups of resources. For instance, in many organizations the service desk is recognized as a function or a team of people. As with any team, the service desk team may be involved in the activities of several practices. These may include the service desk, incident management, service request management, problem management, service configuration management, relationship management practices, and others.

This practice guide describes the service desk practice. When ‘service desk teams’, ‘service desk software tools’, or ‘service desk processes’ are discussed, it is clearly indicated.

The service desk practice is involved in all value streams where the service provider communicates with users. It aims to ensure that these communications are effective and convenient for all parties involved.

2.2 Terms and concepts

2.2.1 Communication channels

The service desk practice involves establishing effective and convenient communication channels between users and the service provider. Usually, there are multiple channels and a need for effective channel integration to provide seamless, convenient user experience.

Good communication channels allow users and the service provider to exchange information in a way that is convenient for all parties and ensures the quality of the information. In this context, the term 'convenient' includes the characteristics outlined in Table 2.2.

Table 2.2 Characteristics of communication channels

Characteristics	Explanation
Accessibility	Communication channels should be accessible. This includes language, format, and special features for any user who is impaired, visually or otherwise. Interfaces may require special applications and devices to access the communication channels, as well as special skills.
Assurance	All parties should be assured that communication channels are genuine, secure, and comply with applicable regulations, policies, and rules.
Availability	Communication channels should be available where and when they are needed. Depending on the service, they may include mobile interfaces of various ranges (from organization-only to global coverage) and options for availability time (from certain working hours to continuous).
Contextual intelligence	Wherever possible, communication channels and relevant contextual information should be integrated. This information may include pre-populated contextual data, communication history, user profiles, and so on.
Familiarity	Familiar communication channels can be more convenient than new, unfamiliar ones. Social media, forums, email, chats, and other communication channels may be effectively adapted for contacts with the service provider.
Integration	Service providers often use multiple channels to communicate with users. Additionally, multiple other systems may be involved in service interactions. These systems should be integrated to reduce or eliminate duplication of data entry and to prevent information loss (see the definition of omnichannel communications below).
Usability	Interfaces of all kinds should be clear, intuitive, helpful, and functional.

The characteristics outlined in Table 2.2 are similar to the characteristics usually used to assess and manage the quality of information, such as: availability, reliability, accessibility, timeliness, accuracy, and relevance. It is important to note that information quality depends on communication quality; many information characteristics depend on the information sources and relevant parties.

Multiple channels are often used for communications between a service provider and its users. Multichannel communications may be convenient, but they can also introduce confusion if they are not integrated. The development of multichannel communications to provide seamless experience and information flow has led to omnichannel communications.

Definition: Omnichannel communications

Unified communications across multiple channels based on sharing information across the channels and providing a seamless communication experience.

More information on omnichannel communications is provided in section 2.4.1.

2.2.2 Service empathy

Definition: Service empathy

The ability to recognize, understand, predict, and project the interests, needs, intentions, and experiences of another party in order to establish, maintain, and improve the service relationship.

Service empathy is important for organizations and those involved in service management. It is based on the ability to *understand* other people's feelings, emotions, and needs (cognitive empathy). However, it is important for the mental health of the service desk agents to constrain the *sharing* of the users' feeling and emotions (emotional, or affective, empathy). A service support agent should not share user's frustration, but they should recognize and understand it, demonstrate understanding and readiness to help, and adjust their actions accordingly.

Although automated communication systems can be enhanced with the emerging capabilities of emotional analysis (based on language, voice, and facial expressions), these systems cannot demonstrate empathy. Service empathy is usually fulfilled by human interactions via channels such as chat, video, and voice calls, and through face-to-face meetings.

Service empathy is an important factor of user satisfaction and service provider success. Service empathy should not only apply to the narrow context of user support and related service interactions, it should apply to all service interactions.

2.2.3 User experience and satisfaction

As a communication interface, the service desk practice significantly influences user experience, customer experience, and the overall success of service relationships. Key user satisfaction factors include the effectiveness and convenience of communication channels and interactions.

Definition: Moment of truth

Any episode in which the customer or user encounters an aspect of the organization and gets an impression of the quality of its service. It is the basis for setting and fulfilling client expectations and ultimately achieving client satisfaction.

Many moments of truth in service relationships occur during communications between users and the service provider, and therefore often involve the service desk practice. Although service quality (utility, warranty, and experience of using the service) largely defines user satisfaction, interactions with the service desk, both as a part of normal service consumption and during user support, contribute a lot to the overall user experience and satisfaction.

The service desk practice is also used for collecting information about user satisfaction. Surveys or other satisfaction research tools generally use the communication channels established by this practice. To collect this information effectively, the practice's communication channels should be perceived as trusted, effective, and convenient by the users; if they are not, responses to surveys and other communications will be affected.

2.3 Scope

The scope of the service desk practice includes:

- establishing and maintaining communication channels and interfaces between the service provider and users
- communicating with users
- enabling, logging, and tracking communications between the service provider and users.

There are several activities and areas of responsibility that are not included in the problem management practice, although they are still closely related to problems. These are listed in Table 2.3, along with references to the practice guides in which they can be found.

Table 2.3 Activities related to the service desk practice described in other practice guides

Activity	Practice guide
Incident resolution and management	Incident management
Management and fulfilment of service requests	Service request management
Definition of content, timing, and format of communications between users and the service provider	All practices providing information to or using information from users. These include incident management, problem management, change enablement, release management project management, software development and management, infrastructure and platform management, information security management, and many others
Monitoring of technology and service performance	Monitoring and event management
Management of improvement initiatives	Continual improvement
Communications between the service provider and stakeholders other than users	Relationship management
Maintenance and improvement of the use of information and knowledge	Knowledge management

2.4 Practice success factors

Definition: Practice success factor

A complex functional component of a practice that is required for the practice to fulfil its purpose.

A practice success factor (PSF) is more than a task or activity, as it includes components of all four dimensions of service management. The nature of the activities and resources of PSFs within a practice may differ, but together they ensure that the practice is effective.

The service desk practice includes the following PSFs:

- enabling and continually improving effective, efficient, and convenient communications between the service provider and its users
- enabling the effective integration of user communications into value streams.

2.4.1 Enabling and continually improving effective, efficient, and convenient communications between the service provider and its users

The support channels available to users and customers should be efficient, effective, and convenient. Convenience can be achieved by providing users and customers with channels which meet their needs. Users' needs may change depending on the geographical region, time of day, preferred language, and accessibility requirements. The more convenient the service becomes, the better the user experience will be.

The choice of communication channels and interfaces is defined by multiple factors, including:

- service relationship model
    - internal or external
    - commercial or subsidized
    - mass, out-of-the-box, or tailored
    - corporate or private
    -
- service relationship type
    - basic, cooperative, or partnership
    -
- service users' profile
    - language
    - age
    - social media activity
    - technology use patterns and preferences

    - location
    - culture
    - diversity
- service provider profile
    - location and organizational structure
    - user satisfaction strategy
    - size and variability of the service portfolio
    - technology capabilities and constraints
- external factors affecting the service relationship, including political, economic, social, technological, legal, and environmental factors.

Communication is not simply sending messages. It should not be assumed that a message has been acknowledged and understood. Every recipient may interpret or understand a message differently based on individual circumstances. The sender should ensure that the intended outcome of their message has been achieved. The recipient should check and confirm that they correctly understand the message that they were sent.

When service channels are selected and designed, it is important to consider user readiness for service use and the associated risks and opportunities. Different channels will introduce different challenges; organizations must be prepared to overcome them. Tables 2.4 and 2.5 outlines some of these challenges.

Table 2.4 User-to-human communication channels and associated challenges

Communication channels	Challenges
Voice and video calls	Limited scalability Unstructured information Subjective attitudes and emotions
Live chat	Limited accessibility Limited scalability Unstructured information Subjective attitudes and emotions
E-mail	Limited accessibility Delayed response Gaps in communication flow Unstructured information
Walk-in	Incomplete or delayed recording Limited accessibility Very limited scalability Unstructured information Subjective attitudes and emotions
Social media and messengers	Viral effect, high exposure of mistakes and conflicts Subjective attitudes and emotions Unstructured information Security constraints

These challenges can be addressed with the following solutions:

- investing in support agents' professional development, emotional intelligence, awareness of diverse cultures, and interests
- limiting human support to where it is needed and justified

- leveraging available resources to automate the logging of unstructured information, where appropriate
- promoting self-service where appropriate to increase adoption
- providing clear security parameters and regularly testing their effectiveness.

Table 2.5 User-to-technology communication channels and associated challenges

Communication channels	Challenges
Web portals, interactive voice (phone) menus, mobile applications, chatbots, and so on	Limited range of tasks can be fulfilled by users at their security level Insufficient and inadequate data Insufficient and inadequate user technology skills Lack of service empathy Limited applicability to complicated and complex situations

These challenges can be addressed with the following solutions:

- Assess user skills and available range of support actions before implementing self-help
- Use channels and interfaces familiar to users
- Ensure a high quality of support history data and knowledge
- When using machine learning, ensure a high quality of data and algorithms

In most cases, service providers use multiple channels. It is important to ensure effective integration between the channels; the communications should be omnichannel, not multichannel. A seamless user journey, in which is it possible to switch between channels without losing or corrupting information, facilitates a positive user experience. Multichannel communications without sufficient integration are likely to create confusion and provoke mistakes. Figure 2.1 illustrates how multiple channels can be used for user support.

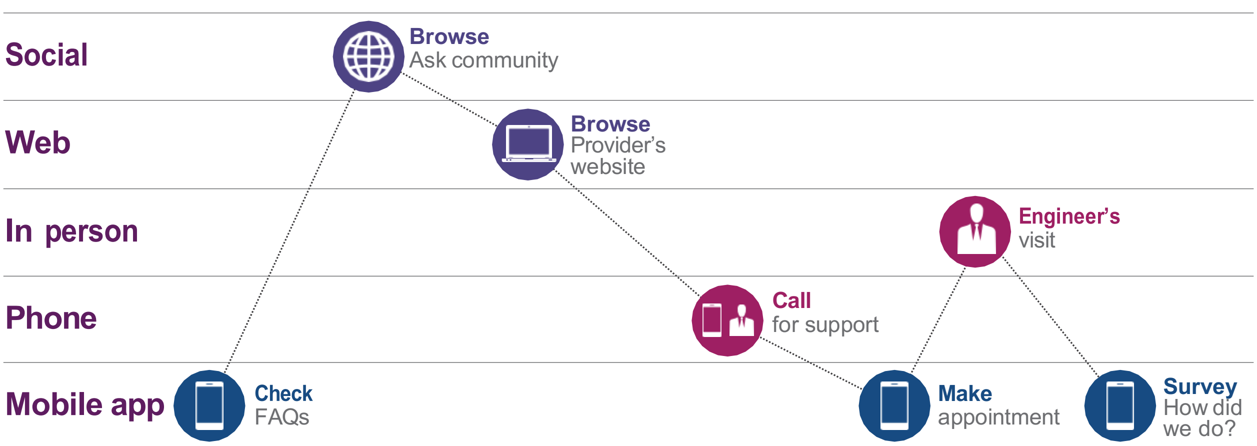

Figure 2.1 Multiple communications channels

In non-integrated multichannel communications, there would be information gaps between the channels; for example, the call for support, appointment in the mobile app, and communications with the visiting engineer all could require re-submitting the request and circumstances that triggered the call for support. Contrary to that, in omnichannel communications, the context would be continually updated, and reusable data would be available wherever relevant. For example, all FAQ search, use of chat bots, and other consultations performed by the user under the same account would add to the context visible to the support specialists. All relevant data would be available for the user support agents and the visiting engineers.

In other words, in multichannel communications, the user would need to start a new journey in every new channel. In omnichannel communications, the journey continues, switching between the channels as convenient.

2.4.2 Enabling the effective integration of user communications into value streams

As a bi-directional communications gateway between the service provider and its users, a key focus of the service desk practice is to effectively capture, record, and integrate communications into relevant value streams. Like most management practices, this practice is involved in multiple value streams: wherever communication between the service provider and its users is needed. See section 3.2 for more on the values streams.

Communications that are initiated by the service provider are defined by and performed in conjunction with one or more other practices involved in the value stream. For example, communications about scheduled changes to services are initiated by and performed in conjunction with the change enablement practice and the release management practice. Communication channels between the service provider and the users are established and managed as part of the service desk practice, but the communication's content, format, and timing are defined as part of the change enablement practice and the release management practice within the context of the value stream.

However, when communications are initiated by users, it is not immediately clear which value stream they belong to, and which ITIL practice activities should be triggered. The service desk practice provides interfaces for communications and procedures for the effective triage of all user queries, including consultations, incidents, service requests, complaints, and compliments. When the user query is triaged and the relevant value stream and practice are identified, the query is processed according to the processes and procedures of the respective practice. In many cases, this involves service desk team resources and/or information systems.

Triage refers to sorting the incoming queue of objects based on their characteristics, urgency, and the likely benefits from processing them. For a service provider, triaging user queries also means categorizing queries and directing them to predefined value streams.

2.5 Key metrics

Key metrics for the service desk practice are mapped to its PSFs. The key metrics are listed in Table 2.6.

The effectiveness and performance of the ITIL practices should be assessed within the context of the value streams to which the practices contribute. The context of the business and the value streams is important when defining whether the practice's performance is considered good or not. Therefore this practice guide cannot recommend universal key performance indicators for the service desk: the target values for each metric can only be defined in the organization's context.

Table 2.6 Key metrics for the service desk practice

Practice success factors	Key metrics
Enabling and continually improving effective, efficient, and convenient communications between the service provider and its users	Quality of the information received via the service desk channels, measured against agreed information quality criteria Convenience of the service desk communications channels and interfaces, measured against agreed convenience criteria Satisfaction of the key communicating stakeholders with the quality of the information and the convenience of the service desk communication channels
Enabling the effective integration of user communications into value streams	Quality of the information received via the service desk channels measured against the requirements of the service value streams (how good is the information captured by the service desk about incidents, requests, and other queries) The satisfaction of the key stakeholders using the information communicated via the service desk channels Number and percentage of the incorrect triage of user queries (queries dispatched to a wrong service value stream/practice/stream)

The correct aggregation of metrics into indicators will make it easier to use the data for the ongoing management of value streams and for the periodic assessment and continual improvement of the service desk practice. There is no single best solution. Performance indicators will be based on the overall service strategy and priorities of an organization, as well as on the goals of the value streams to which the practice contributes.

CHAPTER 3

VALUE STREAMS AND PROCESSES

3 Value streams and processes

3.1 Processes

Each practice may include one or more processes and activities that may be necessary to fulfil the purpose of that practice.

Definition: Process

A set of interrelated or interacting activities that transform inputs into outputs. A process takes one or more defined inputs and turns them into defined outputs. Processes define the sequence of actions and their dependencies.

Service desk activities form three processes:

- user query handling
- communicating to users
- service desk optimization.

3.1.1 User query handling

This process ensures user queries are captured, validated, and triaged for further processing. It includes the activities listed in Table 3.1 and transforms the inputs into outputs.

Table 3.1 Inputs, activities, and outputs of the user query handling process

Key inputs	Activities	Key outputs
User queries Triage guidelines and procedures Service management records: for example, incident records, change records, problem records, and so on Service configuration information, IT asset information, and other relevant information	Acknowledge and record the user query Validate the user query Triage the user query and initiate the appropriate activities	Recorded and categorized user queries Initiated processing of the categorized user queries

Figure 3.1 shows a workflow diagram of the process.

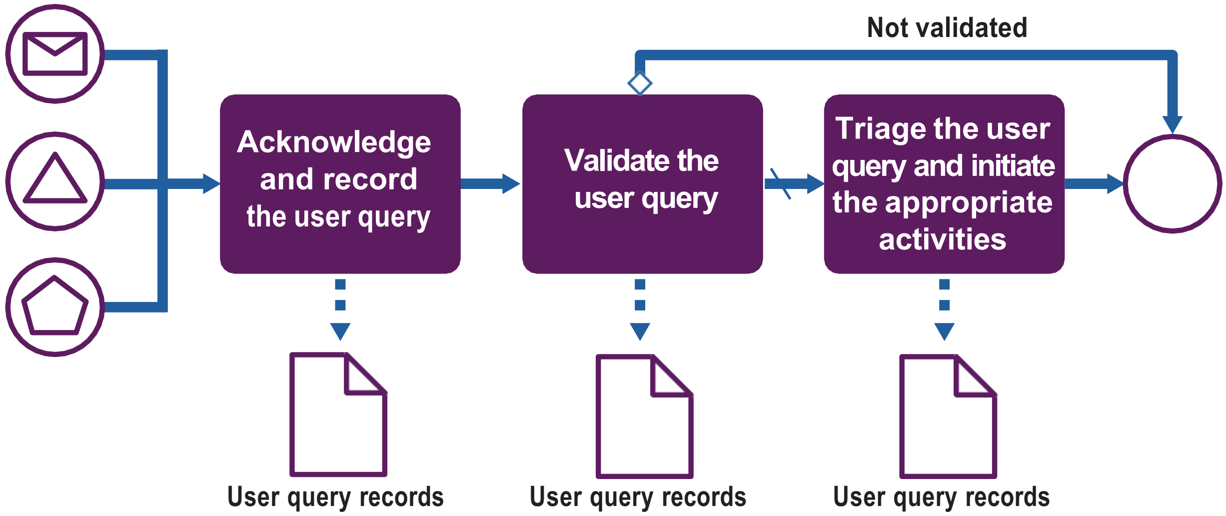

Figure 3.1 Workflow of the user query handling process

Table 3.2 compares human interaction to automation for each of the activities of the user query handling process.

Table 3.2 Human interaction compared to automation

Activity	Human interaction with service desk team	Self-service with automation
Acknowledge and record the user query	Users expect a rapid response if they address the service provider for any reason. Although there are increasing numbers of alternative and sometimes more efficient ways to help users, traditional phone support, email, chat, and walk-in channels remain the key main channels for many organizations, especially between an internal service provider and a wider business organization. Human interaction also enables empathy and relief in purely technological or B2B service delivery environments. Automated support aside, any query that reaches a service desk agent should be met in a polite and standard manner, so that users are met with certain level of quality and shown that their query is welcomed by the service provider. Every interaction must be recorded (meaning uniquely identified in a query log or a user query management and workflow tool). The service desk agents may need incentives encouraging the query recording. Records are an invaluable source of data on service quality, and automation is key to enabling it.	Before the user needs a human response, there can be preliminary stages in response to a query that aim to resolve the query quickly. These are commonly known as self-service tools. For example, when a user contacts the service provider using a chatbot or a phone call to an interactive voice response (IVR), the system usually: • Presents options to categorize the reason for the user's call. This can both automate the query categorization and suggest known resolutions to the user • Publishes important announcements about ongoing service downtime or upcoming changes that are affecting users • Validates the user's identity Apart from these, self-service systems are used to record the query and to provide the user with a confirmation of the query registration.

Table continues

Table 3.2 *continued*

Activity	Human interaction with service desk team	Self-service with automation
Validate the user query.	A service desk agent can perform query validation when recording the query. There are different checks applicable to certain types of queries: • Whether the user is the person they claim to be • Whether the user and their organization are eligible to consume the specified service. This is especially important in commercial service provision, where requested services are chargeable and may be subject to fraud • Whether the reason for the query pertains to the service in question; is the responsibility of the service provider; is included in the agreed level of service, and so on • Whether any protected information needs to be communicated to process the query and, if so, whether additional caller identity checks are required Although data sources, such as service catalogue or identity management systems, enable these checks, the service desk agent is responsible for validating the query.	When self-service tools are employed to query, some aspects of the query validation become automatic. Automated validations of the user and the query can be built into the user journey to enhance and customize it and to prevent questions and option which are not relevant for the user. For example, if a user passes the authorization and authentication checks with the self-service interface (mobile app, web portal, chat, IVR, and so on), the system could match their record against the service catalogue and the knowledge base, and present them with applicable service offerings and recommendations based on eligibility, role, geographic location, service configuration, and so on
Triage the user query and initiate the appropriate activities	For a service provider, triage of the user queries means categorizing and directing them to predefined value streams. *Note: The service desk staff may be involved in the activities of these value streams (for example, classifying an incident, or fulfilling a service request). However, these activities are in scope of other practices, where service desk agents and tools may be involved as resources. See section 3.2 for more on integration of service desk in various service value streams.* Triage of some basic queries may result in a service desk agent resolving them at the first take, as the dialogue with the user progresses. It is important to carefully balance the service desk staff availability to process the incoming queries and their ability to apply the technical competence, especially to time-intensive tasks.	Automation of user query handling ensures that there will be an impartial record of interactions. This can prove useful even for the basic improvement and optimization activities, such as estimating overall demand for user support or calculating the ratio of unaddressed calls. Automated query categorization based on the information collected at the previous steps can reduce human effort and time spent on triaging and routing queries. The interface and workflow of the triage tools should provide a positive user experience and encourage users to communicate their queries. A complicated design with irrelevant or repeating questions may prevent users from contacting the service desk in the future. Using automation, some query types can be resolved with no human interaction (for example, by suggesting a self-help guide or diagnostics steps to the user based on contextual analysis of the query) or by a minimum amount of human interaction.

3.1.2 Communicating to users

This process ensures that various types of information are communicated to users through the appropriate channel(s). It includes the activities listed in Table 3.3 and transforms the inputs into outputs.

Table 3.3 Inputs, activities, and outputs of the communicating to users process

Key inputs	Activities	Key outputs
Requirements for the user communication Information to communicate Records of previous communications Service management records: for example, incident records, change records, problem records, and so on Service configuration information, IT Asset information, and other relevant information	Identifying and confirming the target audience Identifying and confirming communication channels Information packaging Information sending Gathering and processing receipt confirmations and the feedback	Communicated messages Communication reports

All communication between the service provider and the users is made via this process of the service desk practice. Requirements for the communications to users are usually established by various practices. The communications are often standardized and automated: for example, a notification about an incident status change.

Like the processes described in sections 3.1.1 and 3.1.2, this process is largely based on inputs from other practices. Requirements for the user communications (target audience, means of communications, frequency, language, and so on) come from many practices in the context of various service value streams. The same practices usually provide information to be communicated. Service desk practice may employ various means of communications; the channels used to receive user's queries may be amended by others, such as websites, social media, in-app push notifications, and others. Figure 3.2 shows a workflow diagram of the process.

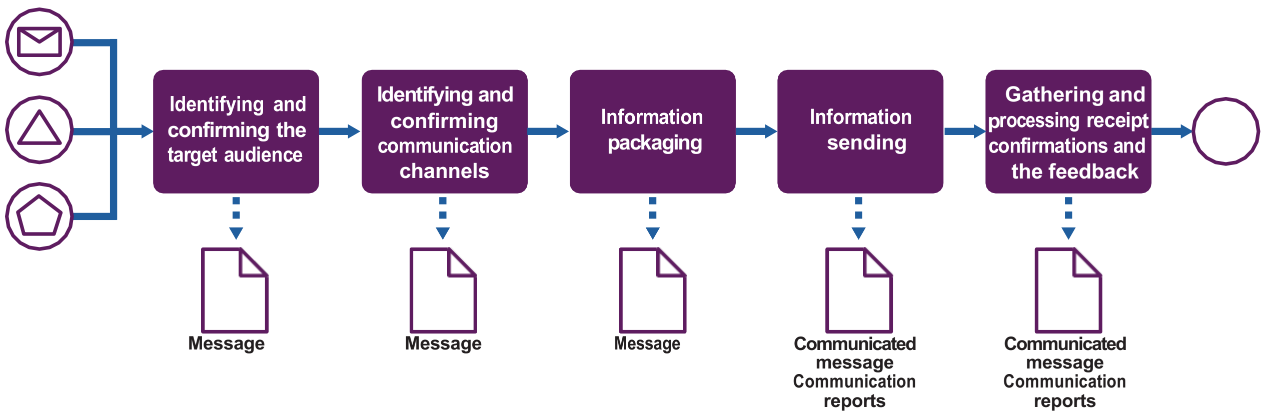

Figure 3.2 Workflow of the communicating to users process

Table 3.4 outlines the activities of the process for different scenarios.

Table 3.4 Activities of the communicating to users process

Activity	Individual tailored communication about previously registered query	Mass communication/broadcasting
Identifying and confirming the target audience	Every outgoing interaction from the service desk must comply with a consistent standard of quality maintained by the service provider, no matter how small the target audience is. Even a query record status update is an outgoing communication that should be carefully designed. Depending on the nature of the query, there can be more than a single recipient of the message, such as stakeholders or service provider staff. Most of the time, a user query management and workflow tool would keep track of the recipients for each user query, and the service desk practice would provide input to design this feature.	There are situations when information needs to be shared with a group of users, usually unrelated to any specific query. This can be a major incident impact or resolution notification, an upcoming service outage related to a change or an emergency, an annual user satisfaction survey, and so on. No matter which value stream the need for mass communication comes from, the service desk practice is used to maintain the standards of communication, perform quality control over the communicated information, and gather feedback on behalf of the service provider. For each mass communication scenario, the target audience should be predefined in the context of the service value stream and communicated to the service desk team. In some cases, it may be possible to confirm the audience with a service provider authority (examples include communications about major incidents and disasters, new releases, changes in terms of service, and other sensitive or important information).
Identifying and confirming communication channels	In an omnichannel paradigm, the user should be able to decide which channel the service provider should deliver information from. Sometimes the preferred communication channels are defined by the customer, agreed in a service level agreement (SLA), and apply to all users covered by the SLA. In other cases, the user can choose for each query how they prefer to be updated on the query status and contacted if any additional information is needed by the service provider.	Means of mass communication should be pre-agreed and tested for every scenario. This is done in the context of the service value streams. Requirements and constraints may come from various practices and should be confirmed based on the service desk capabilities. For every mass communication, there still can be several options available; the final choice of the channel should be confirmed by the communication initiator and in some cases approved by a relevant service provider authority. *Note: In B2B and internal service relationships mass communications via service desk do not include marketing communications to service customers and sponsors. These audiences and messages are beyond the scope of the service desk practice.* *In B2C service relationships, where the roles of sponsor, customer, and user may be combined in one person, the service desk practice may be engaged to deliver such messages.*
Information packaging	For service value streams related to user queries, a set of templates usually exists for all notification types generated over the lifecycle of a query record. It is important to maintain these templates and keep themup-to-date, relevant, and informative. Service provider should encourage and act upon any feedback from users regarding the format of the communications. This feedback serves as input for continual improvement of both service desk practice and the respective service value streams. Custom manual communications beyond the query lifecycle updates should also follow an agreed template and clearly state the purpose of communication, related query records and content.	The agreed templates are applied to the information provided by the sender. Service desk ensures that the agreed templates and procedures are thoroughly followed. It may include approval by pre-agreed service provider authorities, testing of the communication channels and other steps to ensure timeliness, correctness, and delivery to the correct audience. For example, the “WEBAPPS_SRV01 is going down for core patching on Saturday night” is much less preferable from a user’s point of view than, for example: “We are working on improvements to our systems this weekend. Please expect web banking to be unavailable from 6pm Saturday to 12pm Sunday. Our new mobile banking app will work as usual. Thank you for your patience!”

Table continues

Table 3.4 *continued*

Activity	Individual tailored communication about previously registered query	Mass communication/broadcasting
Information sending	Status updates and other communications not requiring a response from the user are typically, sent automatically using the agreed communication channel. Communications requiring a response from the user are often sent by a service desk agent, using an agreed channel allowing a real time dialogue. This may be a chat, a voice or video call, or, less often, an email. Email can be used when no other channels are available or when longer delays in responses are acceptable.	The service desk practice can also have a first-hand knowledge of users' culture that would allow them to choose an appropriate schedule of communication and delivery method. There can be a final approval procedure for certain types of communication, and usually the service desk manager or the role of equivalent authority can issue the message in the name of service provider's service desk.
Gathering and processing receipt confirmations and the feedback	Although many query status updates are sent automatically and do not ask for any specific response from the user, they should provide a clear way to contact the service provider should the user need to. This is particularly important for queries of high urgency and/or business impact, such as incidents. When a final status update is sent to a user (typically, to confirm the completion of the query processing) it often includes a short satisfaction survey. It is good practice to include a question about user's satisfaction with the communications and possible improvements to the service provider's service desk.	Each mass communication needs to have a clear reference to the feedback channel the users can use for this message. This channel is likely to lead back to the service desk, and it is crucial that incoming queries relating to a specific mass communication are identified, recorded, and acted upon, usually by the initiator of the communication. Failure to process mass communication feedback can lead to a decrease of credibility and user attention to the communications from the service provider.

3.1.3 Service desk optimization

This process ensures that the lessons from managing user communications are learned and that approaches to this practice are continually improved. It includes the activities listed in Table 3.5 and transforms the inputs into outputs.

Table 3.5 Inputs, activities, and outputs of the service desk optimization process

Key inputs	Activities	Key outputs
Service desk performance reports Satisfaction surveys and other feedback Technology opportunities Incident and service request reports	Service desk review Service desk improvement initiation Service desk improvement communication	Improvement initiatives Service desk improvement communications

Figure 3.3 shows a workflow diagram of the process.

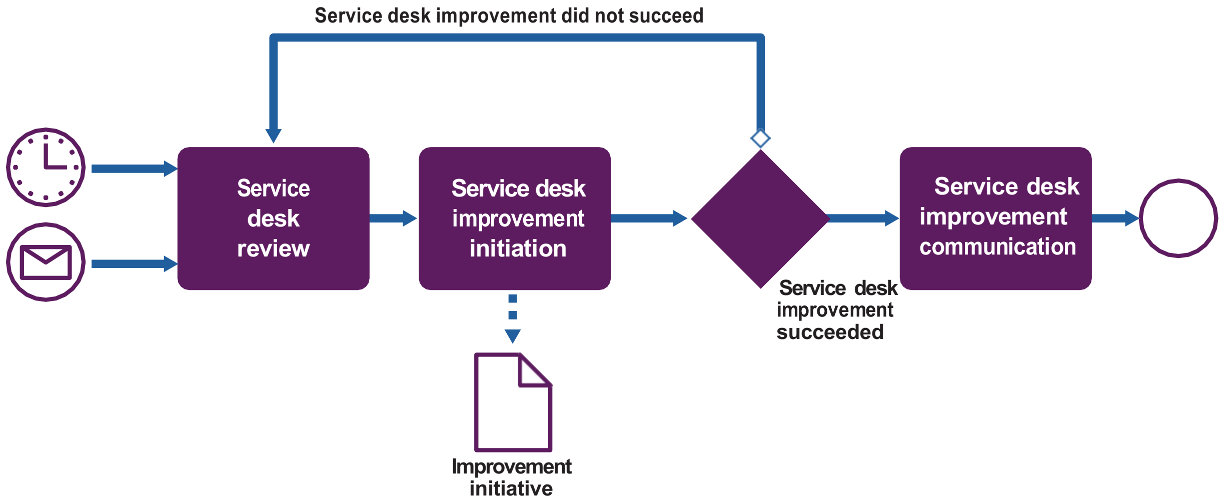

Figure 3.3 Workflow of the service desk optimization process

Table 3.6 outlines the activities of the process.

Table 3.6 Activities of the service desk optimization process

Activity	Description
Service desk review	The service desk team manager together with other relevant stakeholders perform a review of the feedback, performance reports, technology opportunities and other relevant information. The reviews are conducted regularly (usually, monthly or quarterly), or as a reaction to significant events, such as service desk performance deviations, organizational changes, major incidents, or disasters. The participants of the review identify opportunities for the service desk practice's improvement.

Service desk improvement initiation	Depending on the nature of the improvement, the service desk team manager:
	• registers improvement initiatives following a procedure defined in the continual improvement practice:
	  ◦ improvements within the scope of the service desk practice are coordinated by the service desk manager
	  ◦ improvements outside of the practice scope are coordinated by a relevant manager or team; the service desk manager may be informed or consulted during the improvement implementation.
	• initiates a change request.

Service desk improvement communication	Status and progress of the service desk improvements are communicated to the relevant stakeholders. If the improvements have an impact on users, the process of communicating to users (see 3.1.2) is used to send the update.

3.2 Value stream contribution

3.2.1 Service value streams

To perform certain tasks or respond to specific situations, organizations create service value streams. These are specific combinations of activities and practices, and each one is designed for a particular scenario. Once designed, value streams should be subject to continual improvement.

Definition: Value stream

A series of steps an organization undertakes to create and deliver products and services to consumers.

In practice, however, many organizations come to use of the value stream concept after having worked for a while, sometimes for years, without the value streams being managed, mapped, or understood. This means that when the importance of the concept becomes clear, the first step is to understand and map the ‘As Is’ situation, the real flows of work, and to analyse them in order to identify and eliminate the non-value-adding activities and other forms of waste.

Identifying and understanding the existing value streams is critical to improving organization’s performance. Structuring the organization’s activities in the form of value streams allows it to have a clear picture of what it delivers and how, and to make continual improvements to its services. Combined, organizations’ value streams form an operating model which can be used to understand and improve how the organization creates value for the stakeholders.

Many organizations follow best practice recommendations for various service management practices, such as incident management, change enablement, software development, and many others. Problem management is one of the most adopted and developed practices.

However, the practices have often been adopted and organized in a siloed, isolated manner, just as they were presented in the service management bodies of knowledge. In reality, a flow of work required to create or restore value for a customer or another stakeholder is almost never limited to one practice.

3.2.2 The service desk practice in service value streams

The service desk practice is involved in many service value streams. The most obvious two are the restoration of normal operations in case of an incident and the fulfilment of users’ requests. These service value streams are often initiated by user queries, and the service desk practice plays an important role in the initial handling of the queries, their triage and registration. In many cases, the service desk practice is also used to confirm the incident resolution or the request fulfilment with users prior to the closure of the respective records. Finally, status updates and requests for initial information are sent to users (and their responses received and processed) using the service desk during incident investigation or request fulfilment.

Service desk staff can be involved in other activities of these value streams. Service desk agents often perform incident classification; identify and apply known incident resolutions; partially or wholly fulfil service requests. In all these scenarios, the resources of the service desk team are used to perform activities of other practices (incident management and request fulfilment). It is important to distinguish service desk as a practice from service desk as a team.

Key message

The service desk *practice* is a set of capabilities focused on communications with users; the service desk *team* is a team which often has and applies a wider range of capabilities. A service desk team can be involved in any service value stream and in any role where their skills and competencies are needed. The service desk practice can be involved in any value stream where its capabilities are needed.

Other service value streams where the service desk practice is involved, are listed in table 3.7.

Table 3.7 Role of the service desk practice in service value streams

Service value stream	Role of the service desk practice
Development of new and changed services	Informing users about the upcoming and recent updates to the services Collecting feedback about the new and changed services
Ongoing operations of the technology infrastructure	Informing users about service maintenance and its impact on service performance and availability
Onboarding new users to the organization	Informing the new users about the procedures and channels for interacting with the service provider
Development and improvement of the service portfolio	Collecting feedback about the currently available and demanded services

3.2.3 Analysing a service value stream

3.2.3.1 The key steps of a service value stream analysis

The following are some simple and practical recommendations for service value stream analysis and mapping.

1. **Identify the scope of the value stream analysis** Value streams can be mapped to a particular product or service or applied to most or all of them. Value streams may differ for different consumers; for example, incidents can be solved and communicated differently for internal and external customers, or for B2B and B2C products, or for services based on products developed inhouse or sourced externally.
2. **Define the purpose of the value stream from the business standpoint** Make sure the stakeholder's concerns are clearly understood, since they are the ones defining value. In the case of service desk, it is usually users who need a convenient interface to communicate with the service provider; however, there are usually other interested parties. For example, internal users may be unable to provide normal service to a business customer because of the incident, and the value of the value stream should be considered from the business perspective, not solely from the user perspective.
3. **Do the service value stream walk** Walk or directly experience the steps and information flow as they go in practice (consider the Lean technique of Gemba walk):
   a. **Identify the workflow steps**
   b. **Collect data as you walk**
   c. **Evaluate the workflow steps** Typically, the criteria for evaluation are:

- value for the stakeholder (does the step add value for the business stakeholder?)
- effectiveness and performance (is the step performed well?)
- availability (are required resources available to execute the step?)
- capacity (are required resources enough?)
- flexibility (are the required resources interchangeable within the step?).

d. **Map the activities and the information flows** In an ideal situation, the flow goes smoothly without delays and pauses, there are no disconnections between the steps, and the workload is level with minimal (and agreed) variation.

e. **Create and review the timeline and resource level** Map out process times and lead times for resources and workload through the workflow steps**.**

4. **Reflect on the value stream map (VSM)** Identify factors that might not have been entirely apparent at first. The information collected is used at this step to find the waste.
5. **Create a 'to be' VSM** This informs and drives improvement. The value stream should be considered holistically to ensure end-to-end efficiency and value creation, not just local improvements.
6. **Using the 'to be' VSM, plan improvements** Refer to the continual improvement practice guide for a practical improvement model.

3.2.3.2 Service desk considerations in a service value stream analysis

To ensure that relevant service desk activities are included in service value streams, the following steps can be added to the above recommendations.

- At the scoping step (1), identify the key user touchpoints and associated user expectations. These are the points where the service desk practice is likely to be involved.
- Make sure the value stream is understood (step 2) from the standpoint of the business, not only of the service provider. Specifically, understand what do the users expect from or how are they affected by the value stream.
- During the service value stream walk (3a), confirm (and sometimes identify new) touchpoints where users interact with the service provider. Many of these touchpoints will not involve the service desk (use of a mobile app or accessing an information system from a laptop; almost any 'normal' use of the IT services). However, even during normal use of services users may need to be informed about upcoming changes or other relevant circumstances; this communication is likely to come via the service desk.
- During the workflow steps evaluation (3c), evaluate the step's impact on the business value. Special attention should be paid to steps with low business value, low performance, and availability or capacity issues. It is not unusual to find steps which serve some internal control or bureaucratic purposes but delay the incident resolution. In the case of the service desk, opportunities for improvement can be found in unclear or excessive communications, overly complicated and inconvenient forms and procedures, and so on.
- At the reflection and planning steps (4-5), ensure that the service desk processes are optimized for business value throughout the stream, not only within the service desk practice.
- Consider including the creation or updating of user communication templates and procedures in the value stream improvement plans (step 6).

CHAPTER 4

ORGANIZATIONS AND PEOPLE

4 Organizations and people

4.1 Roles, competencies, and responsibilities

The practice guides do not describe the practice management roles such as practice owner, practice lead, or practice coach. They focus instead on the specialist roles that are specific to each practice. The structure and naming of each role may differ from organization to organization, so any roles defined in ITIL should not be treated as mandatory, or even recommended.

Remember, roles are not job titles. One person can take on multiple roles and one role can be assigned to multiple people.

Roles are described in the context of processes and activities. Each role is characterized with a competency profile based on the model shown in Table 4.1.

Table 4.1 Competency codes and profiles

Competency code	Competency profile (activities and skills)
L	**Leader** Decision-making, delegating, overseeing other activities, providing incentives and motivation, and evaluating outcomes
A	**Administrator** Assigning and prioritizing tasks, record-keeping, ongoing reporting, and initiating basic improvements
C	**Coordinator/communicator** Coordinating multiple parties, maintaining communication between stakeholders, and running awareness campaigns
M	**Methods and techniques expert** Designing and implementing work techniques, documenting procedures, consulting on processes, work analysis, and continual improvement
T	**Technical expert** Providing technical (subject matter) expertise and conducting expertise-based assignments

4.1.1 Service desk manager and service desk agent

The common roles of the service desk practice are service desk manager and service desk agent.

The service desk manager is responsible for planning, coordination and improvement of the service desk team(s) within the service provider organization. This involves:

- Understanding of the skills and competencies required to fulfil the roles of the team in the service desk and other management practices
- Understanding, planning, and ensuring the capacity and performance of the service desk team
- Onboarding, training, development, and promotion of the service desk agents

- Creating and maintaining a healthy work culture in the service desk team
- Ensuring the workload balance between the service desk activities and involvement of the team in other management practices
- Cooperation with other managers of the service provider in the service value streams
- Review and continual improvement of the service desk practice.

The service desk manager usually also acts as the service desk practice manager, ensuring the practice ongoing performance and continual improvement. In this role, a service desk manager is also responsible for effective automation of the practice, its integration in the service provider's value streams, and effective use of the third-party resources.

The service desk agent, also often called the service desk analyst, is a member of the service desk team. Service desk agents:

- Perform the service desk activities described in section 3
- Act as relationship agents, ensuring a great user experience and high user satisfaction
- Participate in the activities of other practices as required in the context of the service value streams
- Continually develop relevant skills and competencies
- Support the service desk manager in achieving the service desk team's objectives
- Cooperate with other team members in the context of the service value streams.

The responsibilities of the service desk manager and the service desk agents in the service desk practice activities, along with the required skills, are listed in Table 4.2.

Table 4.2 Responsibilities for service desk activities and required skills

Activity	Responsible roles	Competence profile	Specific skills
User query handling process			
Acknowledge and record the user query	Service desk agent	CA	Communication, writing, business, service awareness, and some level of technical skills
Validate the user query	Service desk agent	CA	Understanding of methods for user validation
Triage the user query and initiate the appropriate activities	Service desk agent	ACT	Understanding demand and classifying based on the process rules
Communicating to users process			
Identifying and confirming the target audience	Service desk agent	CM	Understanding the message and communication needs
Identifying and confirming communication channels	Service desk agent	CTM	Understanding the user communication requirements
Information packaging	Service desk agent	CMT	Communication and writing skills Channel technical expertise
Information sending	Service desk agent Service desk manager	AMT	Channel technical expertise
Gathering and processing receipt confirmations and the feedback	Service desk agent Service desk manager	CMA	Feedback tool technical expertise

Table continues

Table 4.2 *continued*

Activity	Responsible roles	Competence profile	Specific skills
Service desk optimization process			
Service desk review	Service desk manager	LM	Decision making, overseeing other activities, and evaluating outcomes
Service desk improvement initiation	Service desk manager	MA	Knowledge of the improvement process
Service desk improvement communication	Service desk manager	CT	Communication skills, technical skills for the usage of the available communication tools

4.2 Organizational structures and teams

The service desk practice usually has a dedicated team that focuses on performing its processes but can be involved in activities of other practices.

Generally, the service desk team is the first line of user support. In addition to communication with the users, this team can apply documented or partially automated techniques to address certain user (and customer) queries. Practices such as incident management, problem management, request fulfilment, and service level management transfer the knowledge of these techniques to the service desk team via knowledge management tools. The additional workload created by executing these techniques should be regularly evaluated against the purpose of the service desk practice and its effect on the service desk team.

When involved in the incident handling and service request handling workflows, members of the service desk team often ensure effective coordination and communication, especially in the tiered support models with multiple transfers of tasks between the teams. Refer to the incident management and service request management practice guides for more details on how the service desk team may be involved in the respective service value streams.

The following section describes different organizational models that can accommodate tasks that are traditionally assigned to service desk team.

4.2.1 Service desk organization models

When a service provider organization is small and dedicated to a limited number of services, the service desk agent role can be shared among staff. However, this is an inefficient way of communicating with users because of the high workload created by the evolving services and products and the limited value that individual user interactions generate.

Even small internal service providers can benefit from dedicated staff addressing user queries, especially where technological and methodical competencies in backstage service components evolve away from end-user service consumption. Service desk staff, as consumer-facing professionals, normally have strong communication skills and a friendly and assuring demeanour. They are also able to quickly shift between tasks and are generally technically competent.

There are several common ways of organizing a dedicated user communication team, which are discussed in sections 4.1.1.1 - 4.1.1.4.

4.2.1.1 Local service desk team

This organizational model works when the service desk is physically capable of managing omnichannel communication. In other words, where the user base is compactly located; for example, in a single office space for all of the customer service staff, a single team is available even for the walk-in channel, and the local service desk organization is applicable.

This advantages of this are:

- Quick and efficient communications within the team and across the service provider organization. The service desk team should, where possible, be physically located between other service provider teams so that it can learn about news and changes quickly.
- Easy human-to-human contact. The service desk team creates trust and presents the service provider as accessible resource.

The challenges of this are:

- Centralized united teams tend to use query automation tools less. Because work is transparent, people ask why it should be recorded. Similarly, the processes and guidelines are communicated and updated orally. This can result in a lack of control over user communications.
- Physical proximity can cause dependencies on specific individuals, rather than specific roles. This risk should be mitigated by procedural controls, but personal relationships can create back- door methods of support and introduce disruptions to service delivery when those individuals leave.

4.2.1.2 Distributed service desk team

This model is similar to the local service desk team model; the user base is spread over multiple locations but there are still physical communication channels between the users and the service desk team. Each geographical area of users is linked to a dedicated service desk team, and each service desk team coordinates their efforts through the application of common communication standards.

The advantages of this are:

- Ability to expand the service provider presence as the customer organization or the number of customers grows, while maintaining presence and communication standards. Service actions are an important component of any service offering; ensuring that service actions are observable is important for maintaining a positive and cooperative reputation.
- Swiftness of reaction to user queries. The distributed service desk organization is most beneficial to users, as they receive the same consistent and swift response to their service queries across all locations.

The challenges of this are:

- Coordination and automation. As teams are distributed, they require a consistent collaboration environment understand current organizational events. All teams require similar and consistent training and controls. Some service providers adapt to workplace commute times (for example, when all team members are within the same metropolitan area) by adopting a single roster for distributed teams to manage demand fluctuations and relieve duties.
- A distributed service desk results in the duplication of expertise and managerial overhead, however much it is automatically coordinated. Generally, the more non-communication tasks are handled by service desk staff (handling model incidents, IMAC requests, or providing support to users), the more duplication of effort can occur among teams. A service provider should critically examine the value

of a distributed service desk (for example, face-to-face interactions) against the coordination challenges and shared costs of a redundant arrangement.

4.2.1.3 Hybrid service desk organization

Most service providers need to choose an optimal point on the spectrum between a local and virtual organizational model. Several known compromises can be made in this trade-off:

- **Concierge service** Instead of having a distributed network of redundant service desk teams, a service provider can decide to have a limited number of staff and service actions delivered on customer sites, with a centralized tracking system for user queries. The concierge service is an arrangement where a small number of service desk agents are present on customer premises to handle walk-in queries during business hours. These agents can autonomously address basic end-user tasks, such as OS glitches and business application support, major announcements, and customer communication (as opposed to user communication), or even minor IMAC queries, if a small stock of minor components (keyboards, batteries, and so on) can be maintained on site. If issues exceed their expertise or if the queue to their bar exceeds a certain threshold, they escalate the query via a known route. These queries are managed at a central virtual location. Concierge-handled queries must be recorded in a central query management tool alongside other queries (phone, online, and so on). This is a reasonable and highly anticipated compromise to a distributed service desk team, aligning closely with digital transformation efforts in the corporate world.
- **Offshore and shared service desk teams** This is a practice inherited from call centres where, depending on the origin of a call, an operator uses different 'playbooks' to process a caller's requests. Some large global service providers and consumer technology vendors create large service desk hubs in low labour cost locations to provide a relatively low-cost service desk practice. Despite the challenges of this highly virtualized approach, the low price can significantly drive demand for this model.

4.2.1.4 Virtual service desk team

This model emerges when a service desk team cannot physically co-locate with the users. This used to be particularly and almost exclusively relevant to commercial providers of mass services, such as internet access providers or user software vendors. However, with the wide adoption of hybrid and home-based work from 2020, many teams, including the service desk team, had to switch to the virtual format.

A virtual service desk team may structurally resemble the local or distributed teams that work together; or it may be a set of individuals that work from their homes using common user query management and workflow tools.

Lack of physical (walk-in) interactions between the virtual service desk agents and users is partially compensated by more advanced communication channels such as video calls and conferences.

The advantages are of the virtual service desk are:

- Less pressure on the team. A technology barrier helps to create cadence and reduces improper communication between parties.
- Lower costs. Reduced or completely removed costs of the service desk location(s) (office) may significantly optimize overall costs of the team and the practice.

The challenges are:

- The service provider must commit to extensive and ongoing investments in automation tools that enable various communication channels and record management. Automated tools (described in section 5.2) should ensure that customers can submit queries quickly and conveniently and find relevant service provider communications easily. Users who seek human-to-human interaction, should be provided with a variety of convenient and highly available tools, such as online chat, video and audio calls.

4.2.2 Service desk sizing

There is no single method of determining how many service desk teams a service provider needs. Analysis can start with a simple mind map of the key factors that influence the workload. The factors include:

- service desk organization type
- the queueing theory or Erlang variables (query arrival rate, acceptable waiting time, dropout rate, length of queue, and so on)
- additional workload generated for the service desk team by other practices (incidents, service requests, surveys, and so on)
- user and customer service level expectations
- acceptable staff turnover rate.

CHAPTER 5

INFORMATION AND TECHNOLOGY

5 Information and technology

5.1 Information exchange

The effectiveness of the service desk practice is based on the quality of the information used. This information includes, but is not limited to, information about:

- users
- services, including service catalogue and request catalogue, and service levels
- knowledge management systems
- planned and executed changes, change schedule, and the possible impacts of changes
- partners and suppliers, including information on the services they provide
- policies and requirements which regulate service provision
- stakeholder satisfaction with the practice.

This information may take various forms. The key inputs and outputs of the practice are listed in section 3.

5.2 Automation and tooling

The service desk practice can significantly benefit from automation. The term automation is used in this and other ITIL publications to refer to the use of digital technology to enable, support, or enhance various activities. This includes, but is not limited to the full automation of activities where technology solutions remove the need for human intervention. Table 5.1 provides a list of the key automation supporting the practice and their most common application.

Table 5.1 Automation solutions for the service desk practice

Automation tools	Application in service desk
Workflow management and collaboration tools including user query ('ticket') management tools	Management of user query records Communications between service desk agent and users Integration of the practices into service value streams Communications between specialists in service value streams
Work planning and prioritization tools	Planning and tracking of improvement initiatives
Analysis and reporting tools	Practice measurement and reporting
Survey tools	Collecting user feedback for service improvement Collection of feedback for practice improvement

Detailed descriptions of how these tools support the practice's activities are outlined in Table 5.2.

Table 5.2 Details of automation of the service desk activities

Process activity	Means of automation	Key functionality	Impact on the effectiveness of the practice
User query handling			
Acknowledge and record the user query	Workflow management and collaboration tools, including user query ('ticket') management tools	Omnichannel ticket recording and processing Intellectual interactive response (chatbots, IVR…)	High
Validate the user query	Workflow management and collaboration tools, including user query ('ticket') management tools	Integration with relevant systems to perform automated validation	High
Triage the user query and initiate the appropriate activities	Workflow management and collaboration tools, including user query ('ticket') management tools	Integration with the workflows of the relevant service value streams (at least, incident resolution and service request fulfilment)	Very high, especially when the number of queries is high
Communicating to users			
Identifying and confirming the target audience	Workflow management and collaboration tools, including user query ('ticket') management tools Survey tools	Detection of location and language preferences Integration with the workflows of the relevant service value streams	High
Identifying and confirming communication channels	Workflow management and collaboration tools, including user query ('ticket') management tools Survey tools	Detecting communication scenarios applicable for that type of communication	Medium
Information packaging	Workflow management and collaboration tools, including user query ('ticket') management tools Survey tools	Formatting of information Templates management	Medium
Information sending	Workflow management and collaboration tools, including user query ('ticket') management tools Survey tools	Communication approval Omnichannel publishing/ sending	Medium
Gathering and processing receipt confirmations and feedback	Workflow management and collaboration tools, including user query ('ticket') management tools Survey tools	Open/read tracking Feedback/rating collection channels	High
Service desk optimization			
Service desk review	Workflow management and collaboration tools Analysis and reporting tools	Remote collaboration Service desk data analysis	Medium
Service desk improvement initiation	Work planning and prioritization tools	Formal registration of the initiatives	Medium
Service desk improvement communication	Workflow management and collaboration tools	Communicating updates to the relevant teams	Medium to high, especially when organization is large, and number of updates is high

5.2.1 Recommendations for automation of service desk

The following recommendations can help when applying automation to the service desk:

- **Design for the value streams** Service desk as point of contact provides interface to various value streams; it has little value when implemented and automated in isolation. Even if at the beginning the integration is limited to handling of incidents and service requests, make sure that wider integration with other practices is supported by the automation system.
- **Make it omnichannel** Ensure that the toolset supports seamless integration with multiple communication channels, including social media, email, phone, in-app chats, corporate collaboration systems and more. It may be not necessary to activate all available channels at once, but make sure that the integration is possible.
- **Ensure integration with knowledge management system** Service desk team is very likely to be involved in multiple value streams, providing consultations and resolutions to the users. Although these activities are out of scope of the service desk practice, they usually involve both service desk team and service desk information system.
- **Pay attention to measurement and reporting from the beginning** Ensure visibility of the service desk management workload, status of the queries, and satisfaction of the users. Make sure that the current status information is available on a dashboard, and analytical reports can be generated and presented.
- **Ensure that self-help capabilities of the system are available and convenient** User-facing interfaces should be clear, easy to use, informative, and customizable to meet the needs of the organization and specifics of different queries.
- **Ensure usability for all** Service provider-facing interfaces should also be friendly, convenient, and efficient. Service desk agents spend all or most of their work time in this system, so it should be convenient to work with.
- **Leverage machine learning capabilities** They can significantly optimize both self-help experience for users and query categorization activity for the service desk agents.

CHAPTER 6

PARTNERS AND SUPPLIERS

6 Partners and suppliers

Very few services are delivered using only an organization's own resources. Most, if not all, depend on other services, often provided by third parties outside the organization (see section 2.4 of the *ITIL Foundation: ITIL 4 Edition* publication for a model of a service relationship). Relationships and dependencies introduced by supporting services are described in the ITIL practice guides for supplier management and service level management.

Partners and suppliers may support the development, management, and execution of the service desk practice. The forms of support include the following:

- **Performing service desk activities** Service desk teams are often considered as a first step in the career ladder of the IT service provider. Together with high workload and sometimes high level of stress, this leads to a high turnover rate in many service desk teams. This, in turn, creates the need for fast onboarding of new service desk agents, supported by well-documented and highly formalized procedures that can be learned and followed after minimal training. All these have made outsourcing and out-staffing of service desk work possible, and many organizations experiment with third-party resources working in the service desk teams.

  The applicability and success of this approach depend on many factors, including:

  - Type of service relationships between the service provider and the service consumer(s)
  - Level of customization of the IT services
  - Formalization of the service value streams and touchpoints
  - Importance of the user experience and satisfaction
  - Level of automation of the service desk activities
  - Sensitivity of the information exchanged between the users and the service provider.
- **Provision of software tools** Most software tools used for service desk work are shared with other practices. Service desk is one of the first management practice to be automated, along with incident management and service request management. This means that automation requirements related to service desk are usually considered when the organization chooses a service management automation system. However, it is important to ensure that the tools used ensure effective integration of practices into service value streams and effective capturing and use of knowledge across the service provider.
- **Consulting and advisory** Specialized suppliers who have developed expertise in service desk can help to establish and develop the practice, to build the team, implement an information system. This is often the first step in developing a service value system, and there are many ITSM consulting and automation specialists offering their help.

CHAPTER 7

CAPABILITY ASSESSMENT AND DEVELOPMENT

7 Capability assessment and development

7.1 The practice capability levels

The practice success factors described in section 2.4 cannot be developed overnight. The ITIL maturity model defines the following capability levels applicable to any management practice:

Level 1 The practice is not well organized; it's performed as initial or intuitive. It may occasionally or partially achieve its purpose through an incomplete set of activities.

Level 2 The practice systematically achieves its purpose through a basic set of activities supported by specialized resources.

Level 3 The practice is well defined and achieves its purpose in an organized way, using dedicated resources and relying on inputs from other practices that are integrated into a service management system.

Level 4 The practice achieves its purpose in a highly organized way, and its performance is continually measured and assessed in the context of the service management system.

Level 5 The practice is continually improving organizational capabilities associated with its purpose.

For each practice, the ITIL maturity model defines criteria for every capability level from level 2 to level 5. These criteria can be used to assess the practice's ability to fulfil its purpose and to contribute to the organization's service value system.

Each criterion is mapped to one of the four dimensions of service management and to the supported capability level. The higher the capability level, the more comprehensive realization of the practice is expected. For example, criteria related to the automation of practices are typically defined at levels 3 or higher because effective automation is only possible if the practice is well defined and organized.

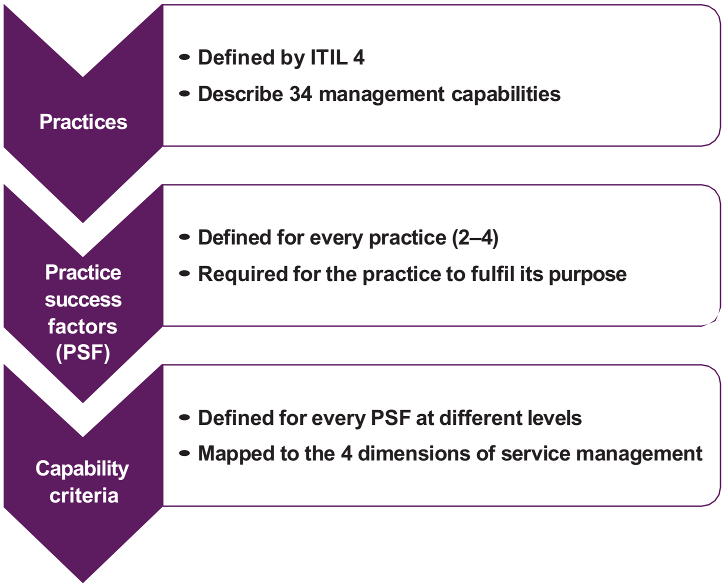

Figure 7.1 Design of the capability criteria

This approach results in every practice having up to 30 capability criteria based on the practice PSFs and mapped to the four dimensions of service management. The number of criteria at each level differs; the four dimensions are comprehensively covered starting from level 3, so this level typically has more criteria than others.

Table 7.1 outlines the capability criteria that are defined in the ITIL maturity model for the service desk practice.

Table 7.1 Service desk capability criteria

PSF	Criterion	Dimension	Capability level
Enabling and continually improving effective, efficient, and convenient communications between the service provider and its users	Channels of communication are established for users to contact the service provider	Value streams and processes	2
	Users are aware of the interface(s) and procedure(s) for contacting the service provider	Value streams and processes	2
	The competencies required to process and manage the users' queries are identified, and qualified human resources are available	Organizations and people	3
	Communication and other technology solutions for processing user queries are identified and implemented	Information and technology	3
	Third party services required to process user queries are identified and available	Partners and suppliers	3
	Policies, templates, and procedures for the management of user queries are agreed and communicated to the relevant members of the organization	Value streams and processes	3
	Contact records and other service desk information are tracked and monitored in an integrated information system	Information and technology	4
	Multiple interfaces for contacting the service provider are unified into an omnichannel solution	Information and technology	4
	The effectiveness, efficiency, and user satisfaction with the point(s) of contact are regularly measured and assessed	Value streams and processes	4
	The effectiveness, efficiency, and user satisfaction with the point(s) of contact are regularly reviewed and continually improved	Value streams and processes	5

Table continues

Table 7.1 *continued*

PSF	Criterion	Dimension	Capability level
Enabling the effective integration of user communications into value streams	User communications in the organization's value streams are performed via the service desk interface(s)	Value streams and processes	2
	Service desk processes and templates are adjusted to meet the needs and context of the value streams	Value streams and processes	3
	Service desk competencies and staffing are adjusted to meet the needs and context of the value streams	Organizations and people	3
	Service desk automation is adjusted to meet the needs and context of the value streams	Information and technology	3
	Service desk use of third-party services is adjusted to meet the needs and context of the value streams	Partners and suppliers	3
	Communications with users are managed in a consistent manner across all value streams	Value streams and processes	4
	Information gathered and used during communications with users is managed in a consistent manner across all value streams	Information and technology	4
	The effectiveness, efficiency, and user satisfaction with the point(s) of contact are measured and assessed in the context of the value streams	Value streams and processes	4
	The integration of user communications into value streams is regularly reviewed and continually improved	Value streams and processes	5

These capability criteria can be used by organizations for self-assessment and improvement of the practice.

7.2 Capability self-assessment

The self-assessment can be conducted by the service provider's internal audit team, if the service provider has one, or by the respective team of the parent organization. If there is no specialized team in the organization, the assessment can be done by a team of practice owners and managers responsible for other management practices of the service provider, or a mixed team of the service provider's executive leaders and managers.

To perform a quick self-assessment using the capability criteria, the following rules should be followed.

1. Start with the level 2 criteria. Based on the knowledge of your organization, answer the question, 'Is this a valid description of our organization in MOST cases?'
2. If the answer to the question above is 'yes', make a list of at least three types of material evidence that could prove the answer. These can be records, documents, interviews with business stakeholders, or service provider's employees.
3. If the answer is 'yes' to all criteria of level 2, this level is considered achieved. Proceed to the criteria of level 3.
4. If not all criteria of level 2 are met, the practice is considered to be at level 1. Focus on the criteria that are not met; what is missing in the organization? Why? How can it affect the quality of the IT products and services? What can be done to meet the criteria that are currently missed?
5. The same approach is applied at every next level; the practice is considered to be at the level, where **all** criteria are met. It is important to focus on the missing capabilities and improvement opportunities, rather than on a formal achievement of a high capability level.

73 Service desk capability development

Management practices should support the achievement of the organization's objectives and enable creation of value for the stakeholders. Depending on the service provider's strategy, positioning, and business and operating models, some practices may be more important and therefore require a higher level of capability, however achieving the highest capability level should never be the aim for all practices. There is no organization that requires all management practices to be at capability level 5. A higher capability level provides higher assurance of the fulfilment of a practice's purpose, but this also comes with an increase in costs (for instance for management, automation, or training). To achieve optimal performance with a sufficient level of assurance, organizations should define a target capability level for each management practice.

Figure 7.2 and table 7.2 show the capability development model, which can be applied to every management practice. The structure of this publication is aligned with the development steps.

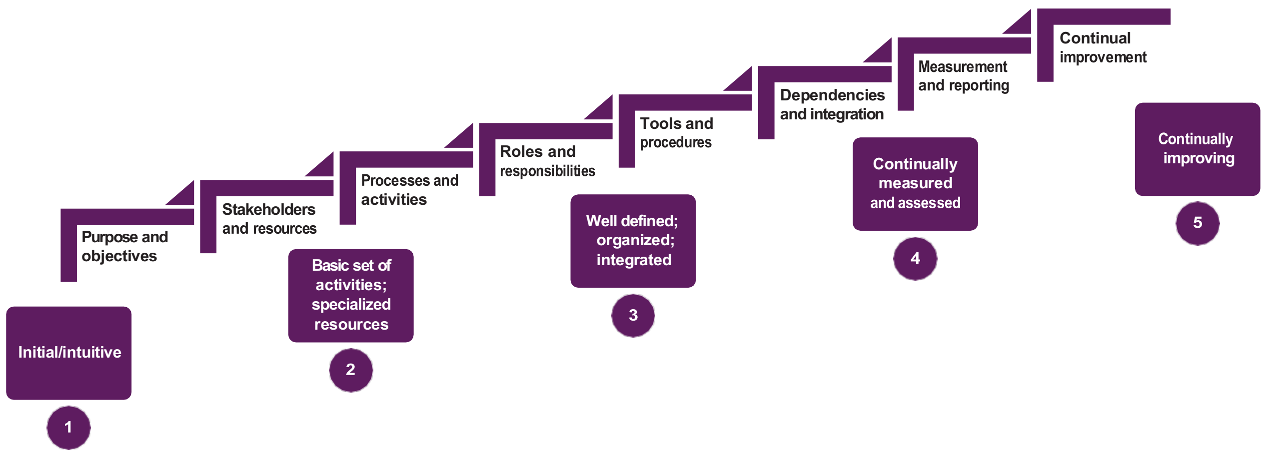

Figure 7.2 The capability development steps and levels

Table 7.2 The service desk capability development steps

Capability level	Define, agree, and implement	Comment for the service desk practice	Chapter (for recommendations)
2	Purpose and objectives	Key types of user queries and of communications with users	2.1
	Scope		2.3
	Processes and activities	Workflows; automated and live communications; query-based and mass communications; roles and responsibilities.	2.2, 3.1
	Roles and responsibilities		4
	Tools and procedures		5
3	Dependencies and integration	Integration in the service value streams	3.2
		Automation and information exchange, use of an integrated service management system	5
		Suppliers and other parties involved in the service desk activities	6
4	Measurement and reporting	Metrics	2.5
5	Continual improvement	Regular review of the practice and the service desk capability development	2.4, 2.5, 7

CHAPTER 8

RECOMMENDATIONS FOR PRACTICE SUCCESS

8 Recommendations for practice success

Most of the content of the practice guides should be taken as a suggestion of areas that an organization might consider when establishing and nurturing their own practices. The practice guides are catalogues of topics that organizations might think about, not a list of answers. When using the content of the ITIL practice guides, organizations should always follow the ITIL guiding principles:

- focus on value
- start where you are
- progress iteratively with feedback
- collaborate and promote visibility
- think and work holistically
- keep it simple and practical
- optimize and automate.

More information on the guiding principles and their application can be found in section 4.3 of *ITIL Foundation: ITIL 4 Edition*.

Table 8.1 outlines recommendations for the success of the service desk practice, linked to the relevant guiding principles.

Table 8.1 Recommendations for the success of service desk

Recommendation	Comments	ITIL guiding principles
Start by establishing a clear communication channel for the users	Do not wait for all possible procedures to be defined, or all possible tools to be implemented. It is more important for the users that you have a friendly interface, and no queries are lost, than some procedures may be imperfect, or some questions may not have immediate answers.	Focus on value Start where you are
Build a team of service desk agents that understand the users' business	Make a tour to the business workplace a part of the service desk agent induction program. Consider training them in the supported business roles. Ensure that the agents understand what the users tell them.	Focus on value Collaborate and promote visibility
Implement new channels and tools gradually, based on the needs and preferences of the users, not only on the technology opportunities. Listen to the users' feedback and improve the channels accordingly	Overcomplicated or unfamiliar interfaces and channels result in low adoption and affect user experience and satisfaction.	Focus on value Progress iteratively with feedback

Table continues

Table 8.1 *continued*

Recommendation	Comments	ITIL guiding principles
Automate repeating and standardized operations	Don't rush to replace human agents with chatbots and other automation. Consider machine learning only when you have sufficient and high-quality data for the learning. Take the users' expectations and habits into account.	Optimize and automate Progress iteratively with feedback
Set objectives and metrics for service desk in the context of the service value streams and of the service consumer's business	Do not focus purely on the service desk performance metrics; understand and support its role in the service provider and the service consumer organizations	Think and work holistically Focus on value Collaborate and promote visibility

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